KIRAN BHATT

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Research Interests

Microbial genomics, antibiotic resistance mechanisms, bioinformatics, One Health, quantitative and data-driven approaches to how microbes respond to their environment

Education

BRAC University, Dhaka, Bangladesh 2020–2024

Bachelor of Science in Biotechnology | CGPA: 3.91/4.00 | Highest Distinction

Degree conferred February 25, 2025

- BRAC University Scholarship: fully funded undergraduate scholarship with stipend throughout degree.
- Undergraduate Thesis: Microbial Analysis of Biomedical Waste under Dr. Munima Haque
- Relevant Coursework: Genomics and Proteomics, Bioinformatics, Advanced Molecular Biology, Microbial Biotechnology, Environmental Biotechnology, Fundamentals of Genetic Engineering, DNA Fingerprinting and Molecular Diagnostics.

Publications

Bhatt, K., Nagarkoti, S., Wazid, M.Z., Haque, M., Hassan, M.M. "Multidrug-Resistant Gram-Negative Pathogens in Biomedical Waste from Tertiary Hospitals in Bangladesh: Phenotypic and Molecular Characterization." Submitted, PLOS ONE.

Research Experience

BRAC University, Department of Biotechnology June 2023-Nov 2024

Undergraduate Researcher, Undergraduate Thesis: Microbial Analysis of Biomedical Waste

- Investigated bacterial pathogens associated with biomedical waste from five hospitals, analyzing 39 samples and 87 bacterial isolates.
- Recovered and characterized presumptive *Pseudomonas, Salmonella*, *Shigella*, and *E. coli* using selective culture, Gram staining, and biochemical testing.
- Performed DNA extraction, species-specific PCR, and agarose gel electrophoresis for molecular confirmation using 16S rRNA, invA, ipaH, and uidA-targeted primers.
- Tested susceptibility to 21 antibiotics using Kirby–Bauer disk diffusion and CLSI-based interpretation, identifying with multidrug resistance in 44.8% of *Pseudomonas* and 91.7% of *Salmonella* isolates.
- Contributed to the analysis of resistance profiles in R using heatmaps, hierarchical clustering and PCA, which showed that *Salmonella* isolates clustered tightly while *E. coli* and *Pseudomonas* isolates were far more diverse.

Bangladesh Council of Scientific and Industrial Research (BCSIR), Dhaka, Bangladesh May 2024

Research Intern

- Performed bacterial culture and isolation, biochemical characterization, DNA extraction, PCR, and agarose gel electrophoresis.
- Applied UV–Vis spectrophotometry and participated in FESEM-based characterization of biological samples.
- Gained practical exposure to plant tissue culture, microbial enzyme assays, probiotic screening, and antimicrobial-resistance analysis using the CARD database.
- Maintained experimental records and followed laboratory documentation and quality-control procedures.

Selected Research & Computational Projects

Equitech Futures, Applied Data Institute| Research Scientist Cohort September–December 2025

Individual Capstone: Healthcare Infrastructure and Antimicrobial Resistance: Multi-Country Analysis of *Klebsiella pneumoniae* Ceftriaxone Resistance Patterns

- Conducted a 41-country ecological analysis examining associations between *K. pneumoniae* ceftriaxone resistance, antibiotic consumption, and healthcare-infrastructure indicators.

- Integrated public AMR, antibiotic-consumption, WASH, healthcare and economic datasets. Applied Ward hierarchical clustering (k=3 by silhouette score), Spearman correlation with bootstrap confidence intervals and multivariable regression.

- Found infrastructure and economic indicators explained more of the variation in resistance (adjusted R^2 0.53) than antibiotic consumption alone (0.11). Limitation: cross-sectional, country-level (ecological) design, so it shows associations, not mechanisms or causation.

Clinical Data Engineering: METABRIC Breast Cancer Dataset June–July 2026

Data Engineering Training Program

- Built a Python/SQL pipeline integrating clinical and molecular breast-cancer data, including tumour characteristics, treatment, relapse, survival, and PAM50 molecular subtypes.
- Designed a Bronze/Silver/Gold data architecture with data-quality checks, relational/warehouse structures, Apache Airflow orchestration, and a Streamlit analytical dashboard.
- Applied explicit handling of incomplete relapse and time-to-relapse data to distinguish outcome reporting from duration-based analysis.

Collaborative Project: Athena Infonomics

- Partnered with Athena Infonomics on a water and sanitation utility-performance dashboard, translating stakeholder requirements into a structured user profile before wireframing.
- Developed the metrics and reporting layer across financial, operational, and service-coverage indicators, contributing from initial wireframe through the final dashboard.

Additional Bioinformatics Projects: Phylogenetic analysis (*PAX5*, multi-species sequence comparison); population-level variant analysis (*DECR1*, gnomAD/ClinVar); protein-protein interaction network analysis (*VHL*, *STAT1*, STRINGdb/Cytoscape)

Research & Computational Training

Stanford Data Ocean 2025

Fundamentals of Data Science in Precision Medicine & Cloud Computing | Fundamentals of AI/ML in Precision Medicine

- Worked with population-scale genomic, cancer, clinical, and multi-omic datasets using Python and SQL.
- Applied statistical and machine-learning approaches to biomedical datasets, including ROC analysis and supervised/unsupervised learning.

Technical Skills

Molecular Microbiology: Bacterial culture and isolation, selective/differential media, Gram staining, biochemical characterization, DNA extraction, PCR, agarose gel electrophoresis, antimicrobial susceptibility testing (Kirby–Bauer, CLSI), aseptic technique

Molecular & Cell Biology: ELISA, SDS-PAGE, Western blotting, brightfield microscopy

Statistics & Data Analysis: Python (Pandas, NumPy, SciPy, scikit-learn, statsmodels), hierarchical clustering, R, PCA, Spearman correlation, bootstrap resampling, regression, data visualisation

Bioinformatics: BLAST, Biopython, multiple-sequence alignment, phylogenetic analysis, gnomAD, ClinVar, COSMIC

Data Engineering: SQL, PostgreSQL, ETL pipeline design, Apache Airflow

Awards & Distinctions

Vice-Chancellor's List, Highest Distinction, BRAC University 2024

BRAC University Scholarship, fully funded undergraduate scholarship with stipend 2020–2024

Languages

English: IELTS Academic 7.5 | Nepali: Fluent | Hindi: Fluent | Bangla: Fluent| French: Basic